

Office hours: 10:00am - 10:50am } MSB 2117
 12:00pm - 1:00pm
<https://www.math.ucdavis.edu/~mgjordan/>

$$4x + 3y = 5 \Rightarrow y = \frac{1}{3}(5 - 4x)$$

$$2x + 7y = 8$$

$$2x + 7\left(\frac{1}{3}(5 - 4x)\right) = 8$$

$$2x + \frac{7}{3}(5 - 4x) = 8$$

$$2x + \frac{35}{3} - \frac{28}{3}x = 8$$

$$\left(\frac{6 - 28}{3}x = \frac{24 - 35}{3}\right) 3$$

$$-22x = -11$$

$$x = \frac{1}{2}$$

$$y = \frac{1}{3}(5 - 4 \cdot \frac{1}{2})$$

$$= \frac{1}{3}(5 - 2)$$

$$= 1 \quad 4x + 3 = 5 \quad x = \frac{1}{2}$$

$$4x + 3y = 5$$

$$(2x + 7y = 8) \times -2$$

$$4x + 3y = 5$$

$$-4x - 14y = -16$$

$$0 - 11y = -11$$

$$y = 1$$

$$\underbrace{3y(5-3x-2y)}_{=0} = 0$$

$$4x(6-2y)=0$$

$$3y = 0$$

~~$y = 0$~~ ①

$$5 - 3x - 2y = 0$$

$$\underline{y = \frac{1}{2}(5 - 3x)} \quad (2)$$

$$4x(x^2 + y^2 - 1) = 0 \quad y = \pm 1$$

$$4y(x^2 + y^2 - 1) = 0 \quad x = \pm 1$$

~~$y = 0$~~

$$\textcircled{1} y=0 \Rightarrow 4x(6-2y)=0$$

$$4x = 0$$

$$x = 0$$

$$(0,0)$$

$$3 \cdot 0 \cdot (5 - 3 \cdot 0 - 2 \cdot 0) = 0 \checkmark$$

$$4.0(6 - 2.0) = 0 \checkmark$$

② $y = \frac{1}{2}(5 - 3x)$

$$4x = 0$$

$$X = 0$$

$$y = \frac{1}{2}(5 - 3 \cdot 0)$$

$$y = \frac{5}{2}$$

$$(0, \frac{5}{2})$$

$$3 \cdot \frac{5}{7} (5 - 3 \cdot 0 - 2 \cdot \frac{5}{2}) = 0$$

$$3. \frac{5}{2} (5-5) = 0 \quad \checkmark$$

$$4 \cdot 0 \cdot (6 - 2 \cdot \frac{5}{2}) = 0 \quad \checkmark$$

$$3 \cdot 3 \left(5 - 3\left(\frac{1}{3}\right) - 2\left(\frac{1}{3}\right) \right) = 0$$

$$4 \left(\frac{1}{3}\right) (6 - 2 \cdot 3) = 0$$

$$6 - 2y = 0$$

$$6 - 2\left(\frac{1}{2}(5 - 3x)\right) = 0$$

$$6 - 5 + 3x = 0$$

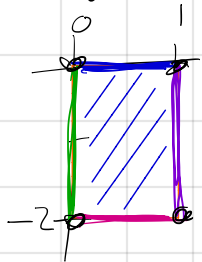
$$x = -\frac{1}{3}$$

$$y = \frac{1}{2} \left(5 - 3 \left(-\frac{1}{3} \right) \right) \\ = \frac{1}{2} (5 + 1) = 3$$

$$\left(-\frac{1}{3}, 3\right) \checkmark$$

10.6

$$17) f(x,y) = x^2 + y^2 - x + 2y \quad 0 \leq x \leq 1 \quad -2 \leq y \leq 0$$



$$\nabla f(x,y) = \begin{bmatrix} 2x-1 \\ 2y+2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2x-1=0$$

$$x = \frac{1}{2}$$

$$\left(\frac{1}{2}, -1\right)$$

$$2y+2=0$$

$$y = -1$$

$$3) x=1$$

$$g_3(y) = (1)^2 + y^2 - (1) + 2y$$

$$= y^2 + 2y$$

$$g_3'(y) = 2y + 2 = 0$$

$$y = -1$$

$$(1, -1)$$

$$1) x=0$$

$$g_1(y) = f(0,y) = 0^2 + y^2 - 0 + 2y$$

$$g_1(y) = y^2 + 2y$$

$$g_1'(y) = 2y + 2 = 0$$

$$y = -1 \quad (0, -1)$$

$$2) y=0$$

$$g_2(x) = x^2 + 0^2 - x + 2 \cdot 0$$

$$g_2(x) = x^2 - x$$

$$g_2'(x) = 2x - 1 \Rightarrow x = \frac{1}{2}$$

$$\left(\frac{1}{2}, 0\right)$$

$$4) g_4(x) = f(x, -2)$$

$$= x^2 + (-2)^2 - x + 2(-2)$$

$$= x^2 - x$$

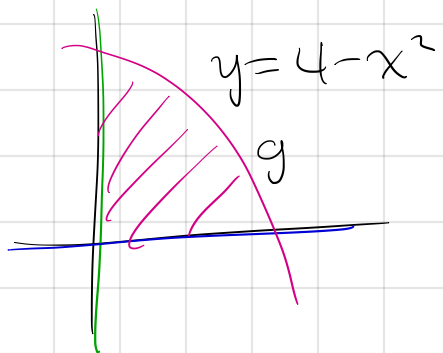
$$g_4'(x) = 2x - 1 \Rightarrow x = \frac{1}{2} \quad \left(\frac{1}{2}, -2\right)$$

type	critical	$f(x,y)$	conclusion
inner	$(\frac{1}{2}, -1)$	$-\frac{1}{4}$	
boundary	$(\frac{1}{2}, -2)$		
	$(\frac{1}{2}, 0)$		
	$(0, -1)$		
	$(1, -1)$		
corner	$(0, 0)$		
	$(1, 0)$		
	$(0, -2)$		
	$(1, -2)$		

$$\left(\frac{1}{2}\right)^2 + (-1)^2 - \left(\frac{1}{2}\right) + 2(-1)$$

$$\frac{1}{4} + 1 - \frac{1}{2} - 2 = -\frac{5}{4}$$

$$-1 - \frac{1}{4}$$



$$g(x) = f(x, 4 - x^2)$$